Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - V

Reinforced Earth and Geosynthetics

Syllabus topics covered in this unit

- Reinforced Earth: Principles
- Components of Reinforced Earth
- Factors Governing Design of Reinforced Earth Walls
- Design Principles of Reinforced Earth Walls
- Geosynthetics: Geotextiles - Types, Functions and Applications
- Geogrids and Geomembranes - Functions and Applications

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Reinforced Earth - Principles

Reinforced earth (reinforced soil) is a composite material formed by the soil (backfill) and the tension-resisting reinforcing elements (strips, grids, sheets, fibres) placed in the soil in layers. It was invented and patented by the French architect-engineer Henri Vidal in 1963 (first wall built in 1965 at Pragnères, France) and named "Terre Armée". In India it is now widely used for flyover approaches, metro viaducts and highway embankments.

Basic principle

- The soil is strong in compression and shear but has no tensile strength. The reinforcement is strong in tension. When the soil mass tries to deform laterally under the load, the friction (or the bearing / the interlocking) between the soil and the reinforcement mobilises the tension in the reinforcement, which restrains the soil - the composite behaves as if it has the cohesion (the apparent cohesion) - "The reinforcement carries the tension and the soil carries the compression".
- The soil-reinforcement interaction is by friction (strips, sheets) and passive resistance / interlocking (grids, anchors).
- The facing (skin) holds the soil at the face and prevents the local sloughing between the reinforcement layers; the real function is that of the soil retention, not the earth pressure resistance - the reinforced mass acts as a gravity wall.

Two views of reinforcement action

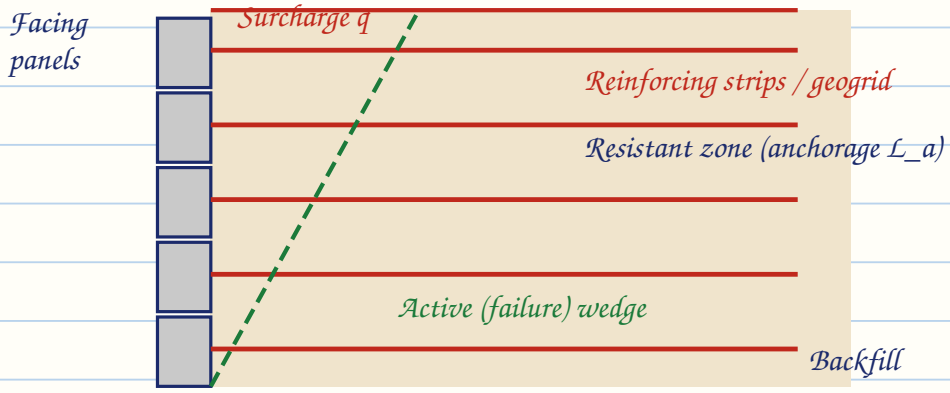
Concept	Explanation
Internal confinement (Vidal) - Apparent cohesion (Schlosser and Long)	The reinforcement restrains the lateral strain of the soil; the soil mass has the anisotropic apparent cohesion: $c_r = T_R \, \mathrm{sqrt}(K_p) / (2 S_v S_h)$, where T_R = the tensile strength of the reinforcement and S_v, S_h are the spacing
Enhanced confining stress (Mohr circle)	The reinforced soil shows the higher strength since the reinforcement raises the minor principal stress: $\mathrm{sigma}_1 = \mathrm{sigma}_3 K_p + 2 c_r \, \mathrm{sqrt}(K_p)$; the failure envelope shifts upward

Advantages of reinforced earth

- Flexible, tolerates the large differential settlement; the quick construction (precast facing); the economical for the height > 4 - 6 m (30 - 50 % saving on the RC walls); the ductile - excellent in the earthquakes (performed well in the Kobe, Northridge); the aesthetics (the precast panels); the less foundation preparation - the walls up to the 20 - 30 m high.
- Limitations: needs the good granular backfill and the space (the reinforcement length 0.7 H); the durability of steel in the aggressive soil; the specialised design and the drainage care; the not suited to the walls in the water (the scour).

Applications

- Retaining walls at the bridge abutments, the approach embankments, the flyovers and the metro; the steep slopes (the reinforced soil slopes - RSS, 1 H : 2 V to 1H : 1V); the embankments on the soft soil (the basal reinforcement); the foundations (the reinforced soil footing - the bearing capacity); the pavement sub-base, the landslide repair, the dams / the dykes, the mining, the noise barrier mounds, the anchors.



2. Components of Reinforced Earth

A reinforced earth wall consists of four main components: the backfill soil, the reinforcement, the facing and the foundation + drainage.

1. Backfill (the fill soil)

- The frictional (cohesionless), the free draining granular soil is the best: the sand, gravel, crushed rock; the angle of friction phi >= 30 degrees (the typical 34 - 36); the fines (< 75 micron) not more than 15 % (FHWA: < 15 %; the PI < 6, the clay content < 10 %); the maximum size 75 - 100

mm (but the 19 mm to avoid the damage of the geosynthetic); the coefficient of the uniformity C_u > 2 - 4.

- Chemical limits for the steel reinforcement: pH 5 - 10, the resistivity > 3000 ohm-cm, the chlorides < 100 ppm, the sulphates < 200 ppm (the organic content < 1 %); the soil with the low pH 4.5 - 9 for the geosynthetics (the polyester pH 3 - 9, the polypropylene / polyethylene are the more resistant).
- The fine-grained and the cohesive soils are used with the geosynthetic (a drainage geocomposite) with the care (the pore pressure).

2. Reinforcement

Type	Description	Features
Metallic strips	Galvanised steel strips - ribbed, 50 - 120 mm wide x 3 - 5 mm thick (typical 50 x 4 mm), the grade Fe 410 / 500; stainless steel and aluminium for special	High tensile strength (up to 400 MPa), the stiff, low strain; the corrosion needs the sacrificial thickness (galvanising 0.086 mm, the loss 0.015 mm / year first 2 years, then 0.004 mm/yr)
Metallic grids / bar mats / welded wire mesh	The steel bar mat (8 - 10 mm bars), the wire mesh	Passive resistance via the cross bars (the more pullout resistance)
Geogrids (polymeric)	The HDPE / polyester (PET) / PP grids, the uniaxial	The non-corroding, the flexible, the interlocking; creep (reduced strength factor)
Geotextiles (woven)	The woven polyester or PP sheets (strength 20 - 400 kN/m)	The low cost; the friction; used for the slopes
Geocells, fibres (the randomly distributed), the soil nails and anchors	The honeycomb, the discrete fibres in the soil (the fibre-reinforced soil), the nails	For the local reinforcement
Natural (jute, coir, bamboo)	Geo-jute, coir geotextile	The temporary / low cost / biodegradable

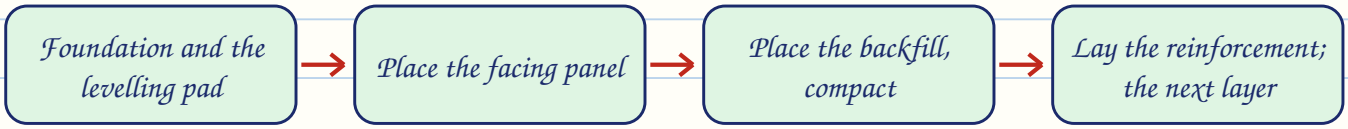
3. Facing (skin element)

- Precast concrete panels (cruciform, 1.5 x 1.5 m, thickness 140 - 180 mm, the hexagonal, the rectangular, the segmental block (interlocking dry-stack blocks), the galvanised steel half-pipe / the sheets (the corrugated, the older type), the welded wire mesh facing (the wrapped geosynthetic - the wrap-around facing), the gabion facing, the shotcrete, the vegetated (green) facing.

- The facing is flexible - the joints with the filter / the geotextile to prevent the loss of the soil (and the neoprene / the polyethylene bearing pads to allow the movement); the facing is connected to the reinforcement by the bolts, the clips, the loops.

4. Foundation and drainage

- A levelling pad (the unreinforced concrete, 150 mm x 300 mm) at the base of the facing; the foundation soil should be able to support the pressure (the bearing, the settlement - the soft soil is improved); the embedment depth = 0.1 H or min 0.5 m (the 1 m).
- Drainage: the drain behind the facing, the weep holes, the geocomposite drain, the blanket drain at the base and the interceptor ditches on the top; the good drainage is essential - no hydrostatic pressure should build up.



3. Factors Governing the Design of Reinforced Earth Walls

Factor	Influence on the design
Wall geometry	The height H, the face batter (the vertical / the inclined), the length of the reinforcement (L / H = 0.7 or more, min 2.5 - 3 m), the top slope (the sloping backfill, the surcharge)
Backfill soil properties	The unit weight gamma, the angle of friction phi (the strength, which determines the K_a, the pullout), the gradation, the drainage, the compaction (95 % MDD), the chemical (corrosion)
Reinforcement properties	The tensile strength T_{allow}, the stiffness (extensible / inextensible), the soil-reinforcement friction coefficient f* or the interface angle delta, the creep, the installation damage, durability and the chemical degradation, the width and the spacing (S_v, S_h), the length L
Loads	The dead loads (the soil weight), the surcharge (the traffic 12 - 20 kPa, the equipment, the buildings), the seismic (the horizontal acceleration k_h), the water, the impact / vehicle collision on the barriers (the parapet)
Foundation soil	The bearing capacity, the settlement (the total and the differential), the slope stability, the ground water; the improvement (the replacement, the stone columns, etc.)
Facing	The type, the connection strength, the flexibility (allowable differential settlement = 1 in 100), the aesthetics
Drainage and the environment	The ground and the surface water, the seepage, the flooding and scour at the toe (the wall in the water), the freeze - thaw
Construction	The compaction near the facing (the light, the 1 m zone), the quality, the construction tolerance, the service life (75 - 120 years for permanent walls: the corrosion allowance), the cost, the availability of the material

- Safety factors (BS 8006 / FHWA / AASHTO / IS 14458 (Part 1 - 3) for the retaining walls in India): the FS against sliding >= 1.5, overturning >= 2.0, the bearing >= 2.5, the pullout >= 1.5 (the reinforcement tension FS 1.5 - 2.5 for the metallic) and the global stability >= 1.3 - 1.5.
- Extensible reinforcement (geosynthetics, a larger strain) vs the inextensible (the steel - the strain less than the soil): they follow the different design approaches (the K-stiffness method and the coherent gravity method).

4. Design Principles of Reinforced Earth Walls

The design of a reinforced earth wall has two parts: the external stability (the whole reinforced mass as a gravity retaining wall) and the internal stability (the reinforcement does not break or pull out). The soil pressure is Rankine active in the retained backfill.

A. External stability (the reinforced block of the width L and the height H)

Checks (the Rankine, coefficient $K_a = \tan^2(45 - \phi/2)$)

- Active thrust of the retained fill: $P_a = 0.5\, K_a \gamma_b H^2$ (+ $K_a q H$ for the surcharge q), acts at H/3.
- FS sliding = $(W \tan \delta_b + c L) / P_a >= 1.5$, $W = \gamma_r H L$ (the weight of the reinforced block); the base friction $\delta_b = \min(\phi_r, \phi_f)$ (the foundation).
- FS overturning = $(W L / 2) / (P_a H / 3) >= 2.0$ (the resultant within the middle third: the eccentricity $e = L/2 - (M_R - M_O)/W < L / 6$ - for the rock L/4).
- Bearing pressure at the base (Meyerhof): $\sigma_v = W / (L - 2e) <= q_{ult} / 2.5$ (the rigid foundation) ; the settlement is checked. Global slip circle analysis (the Bishop) also to be done.

B. Internal stability - the tension and the pull-out (the Rankine method / the Coherent gravity method)

- Lateral earth pressure at the depth z in the reinforced soil: $\sigma_h = K \gamma z$ (+ Kq), $K = K_a$ for the extensible (the geosynthetic - the K constant with depth), K varies from the $K_0 = 1 - \sin\phi$ at the top (z = 0) to K_a at the 6 m depth for the inextensible (steel strips) - the coherent gravity (Juran and Schlosser) method.
- Tension in the reinforcement at the depth z with the vertical spacing S_v and the horizontal spacing S_h: $T = \sigma_h S_v S_h$ (per strip). For the continuous sheets (the per m width): $T = \sigma_h S_v$.
- The failure surface: the Rankine plane at 45 + phi/2 from the horizontal passing through the base of the facing (the inextensible: the maximum tension line - the curved, at 0.3 H from the face at the top to 0.5 (H - z) tan(45 - phi/2) at the base).

Design of the reinforcement

- Tension failure: $T_{max} <= T_{allow} = f_y A_c / FS$ (the net area $A_c = w$ (t - corrosion loss)), $FS_T = T_{allow} / T >= 1.5$ (the steel: 0.55 f_y). For the geosynthetic: $T_{allow} = T_{ult} / (RF_{ID} \times RF_{CR} \times RF_D \times FS)$ with the reduction factors for the installation damage, the creep and the durability.
- Pull-out (the anchorage length L_e) beyond the failure surface: $P_R = 2\ b\ L_e\ sigma_v\ f^*$ (for the strip, the two faces, b = width, f^* = the apparent friction coefficient = tan delta, 1.5 at the top to tan phi at the 6 m depth for the ribbed steel strips; 0.6 - 0.8 tan phi for the geotextile)
- $FS_{pullout} = 2\ b\ L_e\ gamma\ z\ f^* / T >= 1.5 => L_e = FS \times T / (2\ b\ gamma\ z\ f^*)$; Total length $L = L_e + L_r$, where L_r = the length in the active wedge = (H - z) tan(45 - phi/2) (the Rankine).
- Spacing: $S_v = T_{allow} / (sigma_h\ S_h)$ (the typical S_v = 0.3 - 0.75 m for the steel / 0.3 - 0.6 m for the geogrid, the S_h = 0.5 - 1.5 m; the spacing is smaller with the depth).

Fix H, backfill (gamma, phi), the surcharge → Check the external stability (L = 0.7 H) → The tension and the pull-out at each layer → Facing connection, drainage, the global check

Example (the geogrid wall, per m run)

- H = 6 m, gamma = 18 kN/m^3, phi = 32 degree, K_a = 0.307, no surcharge, the S_v = 0.6 m, the geogrid T_{allow} = 30 kN/m.
- At the base layer (z = 5.7 m): $sigma_h = 0.307 \times 18 \times 5.7$ = 31.5 kPa; $T = 31.5 \times 0.6$ = 18.9 kN/m < 30 kN/m (FS = 1.59 > 1.5 OK).
- The anchorage with the tan delta = 0.67 tan 32 = 0.42: $L_e = FS\ T / (2\ gamma\ z\ tan\ delta)$ = 1.5 x 18.9 / (2 x 18 x 5.7 x 0.42) = 0.33 m, therefore the minimum 1.0 m governs; L_r = (6 - 5.7) tan(29) = 0.17 m; top layer L_r = 5.7 x 0.554 = 3.2 m; hence L = 0.7 H = 4.2 m is adopted.

Seismic design

- Mononobe-Okabe the pseudo-static: the dynamic increment $P_{AE} = 0.5\ gamma\ H^2\ (K_{AE})$ and the inertia force of the reinforced block (the 50 % of the dynamic thrust acts at the 0.6 H); the additional tension; the reinforcement length and the pull-out are checked for $k_h = 0.5\ A_{max}$ (the

IS 1893).

5. Geosynthetics - Introduction and Geotextiles

Geosynthetics are the planar, synthetic (polymeric) products - made of the polypropylene (PP), polyester (PET), polyethylene (PE / HDPE), polyvinyl chloride (PVC), the nylon, the polyamide - used with the soil, rock or the other geotechnical material as an integral part of the civil engineering structure. Their use started in the 1960s (the first woven geotextile in the 1958, Florida - the erosion control; Rhone Poulenc 1970s - the non-woven).

Functions of geosynthetics (SFDRC)



Geotextile: S F D R | Geogrid: R | Geomembrane: C (barrier)

Types of the geosynthetics

- Geotextiles (the woven, the non-woven, the knitted), Geogrids, Geomembranes, Geonets, Geosynthetic clay liners (GCL), Geocomposites (the drain), Geocells, Geopipes, Geofoam (EPS blocks), the geo-strips.

Geotextiles - types

Type	Manufacture	Properties
Woven	The weaving of the slit films / the monofilament yarns on a loom (the warp and the weft)	The high strength and low strain (20 - 800 kN/m); low permeability (the less open area); the reinforcement and the separation
Non-woven needle-punched	The fibres (staple or the continuous) are mechanically bonded by the needles	The thick, the high permeability and the flexible, the good in-plane drainage, the filtration, the cushion (the protection), the strength 8 - 30 kN/m, the elongation 50 - 100 %
Non-woven heat-bonded / spun-bonded	The fibres are bonded by the heat (the thermal) or the resin	The thin, the stiff, the filtration
Knitted	The interlocking loops	Special - the high strain
Composite / the geotextile-geogrid	The woven + non-woven	The strength + the filter

Functions of the geotextiles (the 5 functions - SFDRP)

Function	Description	Application
Separation	Keeps the two dissimilar soils (the soft sub-grade and the aggregate base) from mixing; keeps the base thickness	The unpaved and the paved roads, the railways (the ballast), the airfields, the parking
Filtration	Allows the water to pass and retains the soil particles (a replacement of the sand filter)	The drains (the French drain, the blanket / the toe drains), behind the retaining walls, the riprap and the revetments, the dams
Drainage (in-plane)	The transmission of the fluid in the plane of the thick non-woven	The drainage of the pavement, the retaining walls, the landfill leachate
Reinforcement	The tensile strength increases the bearing capacity, the embankment stability	The soft ground embankments, the steep slopes, the walls, the sub-base
Protection (cushion)	The cushioning of the geomembrane against the puncture	Below / above the geomembrane in the landfill, the ponds
Erosion control	Prevents the soil loss on the slope	The coastal, the river bank protection, the silt fence

Properties and tests of the geotextiles

- Physical: the mass per unit area (the g/m^2, 100 - 800), the thickness (IS 13162 Part 3), the apparent opening size (AOS, O_{95}); Mechanical: the wide-width tensile test (IS 13162 Part 5, the

strip 200 mm), the CBR puncture, the trapezoidal tear, the burst; Hydraulic: the permittivity (psi = k_n / t, 1/s), the in-plane transmissivity, the gradient ratio / the clogging; Endurance: the UV resistance, the creep, the chemical and the biological resistance.

- Filter design (the soil retention): O_{95} <= B D_{85} (B = 1 for the coarse; the 0.5 - 1.8) and the permeability $k_{geotextile}$ >= k_{soil}; the clogging (the porosity > 30 %, the AOS > 3 D_{15}).

Applications of the geotextiles

- Roads: the separation + the reinforcement on the soft sub-grade (the reduction of the aggregate thickness 30 - 50 %); the railway track; the airport runway; the asphalt overlay (the paving fabric to stop the reflection cracking).
- Drainage: the filter wrap of the sub-surface drains, the toe filter of the dams, the vertical drain jacket; the reinforced soil (the walls, the slopes, the embankment on the soft clay); the erosion control (the river training, the coastal revetments, the geotube (the sand-filled big tubes for the breakwater and the land reclamation)); the silt fences.

6. Geogrids - Functions and Applications

Geogrids are the open grid-like polymeric structures with the large apertures (the openings 10 - 100 mm) and the tensile ribs (the longitudinal and the transverse) which interlock with the soil / the aggregate particles; their main function is the reinforcement. The geogrid-soil interaction is the friction on the ribs and the passive bearing resistance on the transverse members.

Type	Manufacture	Features
Uniaxial (the tensile strength in one direction)	The extruded HDPE sheet punched and stretched in one direction; or the woven / the knitted PET coated with the PVC / the bitumen; the polyester yarn grids	The strength 40 - 400 kN/m; the low creep; the retaining walls, the steep slopes, the embankments on the soft soil
Biaxial (the same in both)	The punched and stretched in both directions; the PP	The strength 20 - 60 kN/m in both; the road base, the railway sub-ballast, the working platform on the soft ground
Triaxial / Multi-axial	The triangular apertures	The radial stiffness, the good confinement of the aggregate
Welded / the bonded (the flexible)	The polymer straps welded at the junctions	The high junction strength

Functions of the geogrids

- Reinforcement (the main): the tensile strength, the lateral restraint (the confinement of the aggregate that increases the stiffness, the "mechanical stabilisation" and the load spreading in the base layer), the bearing capacity improvement (the tensioned membrane effect).
- Separation and the stabilisation (the minor, the interlock with the aggregate), the strengthening of the asphalt (the grid in the overlay, the cracking control).

Applications

- Reinforced earth walls and steep slopes (with the wrap-around or the modular block facing); the embankment over the soft clay (the basal reinforcement; the stability and the reduced settlement); the bridge abutment, the landslide repair.
- Unpaved / paved roads over the soft sub-grade (the base reinforcement: the thickness reduced up to 40 %, the traffic benefit ratio TBR of 2 - 10 times, the rutting reduced), the railway ballast, the airfield; the foundation on the weak soil (the reinforced sand layer, the bearing capacity ratio BCR 1.5 - 3); the piled embankments (the load transfer platform); the landfill capping and the void bridging (the sinkholes, the mines).
- Geocells (the 3-D honeycomb grids) for the load support, the slope and the channel protection, the geo-webs.

Design strength of the geogrid

- $T_{allow} = T_{ult} / (RF_{CR} \times RF_{ID} \times RF_{D} \times FS)$ (the creep 1.5 - 2.5 for the PP, 1.3 - 1.6 for the PET; the installation damage 1.05 - 1.4; the durability 1.1 - 1.3; FS = 1.3 - 1.5 for the overall uncertainty).
- Example: T_{ult} = 100 kN/m; RF_{CR} = 1.6, RF_{ID} = 1.1, RF_{D} = 1.1, FS = 1.3 => T_{allow} = 100 / (1.6 x 1.1 x 1.1 x 1.3) = 39.7 kN/m.

7. Geomembranes - Functions and Applications

Geomembranes are the continuous, thin, flexible, practically impermeable polymeric sheets (the permeability 10^{-11} to 10^{-13} cm/s; the thickness 0.25 - 3 mm, the rolls 7 - 10 m wide). The main function is the barrier (the containment / the liquid and the vapour barrier).

Material	Features	Typical thickness
HDPE (high density polyethylene)	The most widely used: the high chemical resistance, the strong, the UV resistant (the carbon black), the stiff, the weldable by the thermal fusion / the extrusion; but the stress cracking	1.0 - 2.5 mm
LLDPE / LDPE	More flexible, the better conformability to the deformation	0.5 - 1.5 mm
PVC (polyvinyl chloride)	The flexible, the easy to install, the solvent / the thermal welded; the plasticiser lost (the ageing)	0.5 - 1.5 mm
fPP (flexible polypropylene), EPDM rubber, CSPE (Hypalon), the bituminous geomembrane	The special: the roofing, the tanks, the canals	1 - 2 mm

Functions

- Barrier / the liquid containment (the main): to prevent the flow of the liquid or the gas (the seepage, the leachate, the contaminants); the vapour barrier; the moisture control (the expansive soil - the vertical and the horizontal moisture barrier, the unit I).
- The protection of the soil from the water / the chemical; the separation (the minor).

Applications

- Landfills: the liner and the cover (the composite liner = HDPE geomembrane (1.5 - 2 mm) + the compacted clay liner / the GCL with the leachate collection above); the hazardous waste containment, the mine tailings and the heap leach pads.
- Canals, reservoirs and tanks: the lining to reduce the seepage losses (the irrigation canals - the water saved 20 - 40 %), the farm ponds, the fish ponds, the dams (the upstream face, the core), the tunnel water proofing, the roof gardens.
- Underground storage tanks (the secondary containment), the oil / the chemical storage, the sewage lagoons, the moisture barrier under the pavement and the foundation in the expansive soil, the floating covers (the biogas).
- Installation: the sub-grade preparation, the anchor trench, the seaming (the wedge / the extrusion welding with the quality control by the air pressure and the vacuum box tests), the protection by the non-woven geotextile (the 400 - 800 g/m^2) cushion, the ballast; the wrinkles and the puncture are the major risks; the leak detection (the electrical, the spark test).

Other geosynthetics

- Geonets (the drainage core, the 5 - 10 mm thick, the crossing ribs - the in-plane drainage in the landfill leachate collection); GCL (the bentonite sandwiched between the geotextiles, the $k < 10^{-9}$ cm/s, the self-sealing); geocomposites (the geotextile + the geonet / the grid; the prefabricated vertical drains); the geopipes (the perforated, the corrugated).

Geosynthetic	Main functions	Typical application
Geotextile	S F D R P	Roads, the filter, the drains
Geogrid	R (the confinement)	Walls, the soft soil, the road base
Geomembrane	Barrier	Landfill, the canal and the pond lining
Geonet / composite	D (drainage)	Leachate collection, the sports field
GCL	Barrier	Landfill cap, the liner

Quick revision of Unit V

- RE = soil + the tension elements; the friction; the apparent cohesion; the Vidal 1963. Components: the backfill (phi >= 30, the fines < 15 %), the reinforcement (the steel strips, the geogrid), the facing, the drainage.
- Design: external (the sliding 1.5, the overturning 2, the bearing 2.5), the internal: $T = K \gamma z S_v S_h$, the pull-out $L_e = FS\ T / (2\ b\ \gamma\ z\ f^*)$, $L = 0.7\ H$.
- Geotextile: separation, filtration, drainage, reinforcement, protection. Geogrid: the reinforcement by the interlock. Geomembrane: the barrier (HDPE) for the landfill and the canal lining.

Important Questions (JNTUA pattern)

* Explain the principle of reinforced earth. List the advantages and applications.
* Describe the components of a reinforced earth wall. What are the requirements of the backfill?
* Discuss the factors governing the design of reinforced earth walls.
* Explain the design procedure of a reinforced earth wall - the external and the internal stability. A 6 m high wall with phi = 30 degree and gamma = 18 kN/m^3 is reinforced with the strips at S_v = 0.75 m; find the tension at the base layer.
* What are geotextiles? Explain their types, functions and applications.
* Explain the types, functions and applications of geogrids.
* Describe the geomembranes: the materials, the functions and the applications in landfills and canals.